Immunosuppressive Properties of Methotrexate: Apoptosis and Clonal Deletion of Activated Peripheral T Cells

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Abstract

The folate antagonist methotrexate (MTX) is extensively used in graft-versus-host disease, rheumatoid arthritis, and other chronic inflammatory disorders. In addition to its an-tiinflammatory activity associated with increased release of adenosine, MTX exerts antiproliferative properties by inhi-bition of dihydrofolate reductase and other folate-depen-dent enzymes. However, the mechanisms of immunosup-pressive properties associated with low-dose MTX treatments are still elusive. We report here that MTX (0.1–10 m M) in-duces apoptosis of in vitro activated T cells from human pe-ripheral blood. PBL exposed to MTX for 8 h, then activated in drug-free medium, underwent apoptosis, which was com-pletely abrogated by addition of folinic acid or thymidine. Apoptosis of activated T cells did not require interaction be-tween CD95 (Fas, APO-1) and its ligand, and adenosine re-lease accounted for only a small part of this MTX activity. Apoptosis required progression of activated T cells to the S phase of the cell cycle, as it was prevented by drugs or anti-bodies that interfere with IL-2 synthesis or signaling path-ways. MTX achieved clonal deletion of activated T cells in mixed lymphocyte reactions. Finally, in vitro activation of PBL taken from rheumatoid arthritis patients after MTX injection resulted in apoptosis. Altogether, the data demon-strate that MTX can selectively delete activated peripheral blood T cells by a CD95-independent pathway. This prop-erty could be used as a new pharmacological end point to optimize dosage and timing of MTX administration. It may account for the immunosuppressive effects of low-dose MTX treatments. (*J. Clin. Invest.* 1998. 102:322–328.) Key words: • T • thymidylate synthase • rheumatoid arthritis

Introduction

Methotrexate (MTX) [1] is a folate antagonist first developed for the treatment of malignancies (1) and, subsequently, used in nonneoplastic diseases as an antiinflammatory and/or immunosuppressive drug. MTX is currently the most commonly used treatment of rheumatoid arthirtis (2, 3), and other chronic inflammatory disorders. MTX is also effective in the prophylaxis of acute graft-versus-host disease either alone or in association with cyclosporin A (CsA) and/or prednisone (4–6) or FK506 (7). MTX has also been used as an adjunct therapy for persistent mild cardiac allograft rejection (8). Most pharmacological studies have addressed the use of MTX in cancer chemotherapy, where doses could be escalated up to 30 g/m ^{2} by administration of the antidote leucovorin (folinic acid, the addition g-linked group of the neighboring folyl glutamate, using ATP as its en-ergy source. The activity of this enzyme, first demonstrated in erythrocytes and then in human liver, was shown to vary among tumor cell lines. Surprisingly, the polyglutamation of drug-free medium, thus, providing the first evidence for an immunosuppressive activity of low-dose intermittent MTX administration. Selective susceptibility of actiavted T cells in the S/G2 phase of the cell cycle may result in clonal deletion of T cells that are activated by antigen at the time of MTX adminis-tration.

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1. *Abbreviations used in this* CsA, cyclosporin A; DHFR, dihydrofolate reductase; MLR, mixed lymphocyte reactions; MTX, methotrexate; RPM, rapamycin; TS, thymidylate synthase; TUNEL, Tdt-mediated dUTP-FITC nick end labeling.

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Methods

Reagents and monoclonal MTX, folic acid, folinic adenosine, thymidine, adenosine deaminase a , b-adenosine-5 9-diphosphate (ACPD), PHA, PMA, ionomycin, canavalin A, *Staphylococcus aureus* enterotoxin B were obtained from Sigma Chemical Co. (St. Louis, MO). Rapamycin (RPM) and FK506 were a gift from A. Altman (La Jolla Institute for Allergy and Immunology, La Jolla, CA), and CsA was supplied by Sandoz Phar-maceutical Division (Novartis, Paris, France). The CD3 mAb OKT3 was from Cilag Laboratories (Levallois-Perret, France). The CD25 mAb ARIL-2 (IgG1) and anti-thymocyte globulins were a gift from 7 Alberici (Pasteur-Merieux, Lyon, France). The agonist (CH11, IgM) anti–human CD95 mAb was from Coulter-Immunotech (Marseille, France). The antagonist anti –human CD95 mAb ZB4 (IgG1) was from Kamiya Biomedical (Thousand Oaks, CA). The CD25, CD69, and CD95 mAbs (fluorescein isothiocyanate conjugates) were

Cell preparation and PBL were collected from healthy donors in the presence of sodium citrate. Blood was defibrinated, and then mononuclear cells were isolated by centrifugation on a layer of Histopaque® (Sigma Chemical Co.). Those cell suspensions, referred to as PBL, contained pression. PBL were resuspended in Rosewell Park Memorial Institute culture medium (RPMI 1640; Sigma Chemical Co.), supple-mented with 10% France L-glutamine, and antibiotics (penicillin 100 streptomycin m g/ml). Cultures were maintained at mid atmosphere containing 5% CO_2. During the last 8 h of incubation Int., Little Chalfont, m Ci/well. ^{3}H-TdR sured using a Packard direct beta counter (Meriden, CT) after har-vesting. For mixed lymphocyte reactions (MLR), the human B lym-phoma cell lines RAJI and DAKIKI were used as stimulators (13). Stimulator cells were treated for 1 h at 37

Measurement of nuclear apoptosis. After 3 d of activated PBL were harvested. Dead cells were removed by centrifu-gation on a layer of Histopaque® (Sigma Chemical Co.). Viable cells (10^6/ml) were washed in HBSS, and then incubated in 96-well micro-plates (Costar, Cambridge, MA) with MTX. In other experiments, PBL were either incubated for 1–24 h in the Chemical Co.) at 10 m g/ml after previously Apoptosis was also measured by flow cytometry after addition of biotinylated annexin V (Boehringer Mannheim, Indianapolis, IN; 15) and by TdT-mediated dUTP–FITC nick end labeling (TUNEL), as previously described (16), using reagents from Boehringer Mann-heim. Samples were analyzed by flow cytrometry on a FACScan® (Becton Dickinson). Nuclear fragmentation and/or marked conden-sation of the chromatin with reduction of nuclear size were consid-ered as typical features of apoptotic cells. Based on these measure-ments, results were expressed as percentage of apoptotic cells or percentage of specific apoptosis according to the following formula:

Immunofluorescence assays. After 1 d of culture, washed with isotonic NaCl/Pi buffer containing 1% BSA and 0.2% NaN_3 (PBS/BSA/azide), and then m l of fluorescein isothiocyanate conjugated CD25, CD69, or CD95 mAb $3\ 10^5$ cells for 30 min at 4 8 C. After two washes in PBS/BSA/azide, cell were fixed with 1% formaldehyde in PBS/BSA/azide. Lymphocytes were identified by cell size (small angle light diffraction) and this fraction was gated for analysis, thereby excluding dead cells but including small membrane potential (Dc $3\ 10^5$/ml) were incubated with 3,3 9-dihexyloxacarbocyanine ($DiOC_6(3)$) 40 nM in PBS (Molecular Probes, Inc., Eugene, OR) for 15 min at 37 8 C, followed by flow cy-tometry analysis (l . Ex. Max., 488 nM; l . Em. Max., 525 nm). The de-crease in Dc m is a characteristic of apoptotic cells (18).

CD95-ligand mRNA quantification was carried out following methods described in detail in another report (19).

Patients. Peripheral blood from one patient with a defect in the CD95 apoptotic signaling pathway was kindly made available by Alain Fischer (Hôpital des Enfants Malades, Paris, France). Details of the phenotypic alterations of this patient have been previously re-ported (20). Six patients with rheumatoid arthritis treated by weekly intramuscular injections of MTX (7.5 mg in one patient, 12.5–15 mg in the five others) in the clinical immunology unit, Hôpital E. Herriot (patients of P. Miossec) gave their informed consent for blood sam-pling (5 ml) before, and 6, 24, and 48 h after

Results

MTX induces apoptosis of mitogen-activated but not resting pe-ripheral T cells. PBL were activated by PHA for 3 d, and then dead cells were removed by centrifugation, and viable cells were incubated for 15 h with MTX (1 m M). Apoptosis was demonstrated by internucleosomal fragmentation resulting in a typical "ladder" of 180-bp fragments and multiples thereof in agarose gel electrophoresis, whereas such fragments were not detected in nonactivated PBL (Fig. 1 *A*). Apoptosis of PHA-activated cells, but not resting PBL was confirmed by typical condensation or fragmentation of cell nuclei, as revealed by Hoescht 33342 staining in fluorescence microscopy (Fig. 1*B*), electron microscopy (Fig. 1 *C*), by the decrease of mito-chondrial transmembrane potential (Dc m) (Fig. 1 *D*), and the presence of DNA breaks revealed by the TUNEL assay, (Fig. 1 *D*).

The kinetics of apoptosis showed a progressive increase up to 16 h (10 m M) or 28 h (0.1 m M) (data not shown). Similar ob-servations were made with PBL activated by PMA plus iono-mycin, concanavalin A, the CD3 mAb OKT3, polyclonal rab-bit anti-thymocyte globulins and *S. aureus* enterotoxin B, but not by PMA alone (data not shown). We concluded that MTX (1 m M) induced apoptosis of activated T cells. A dose range of MTX concentrations was tested, demonstrating that apoptosis occurred at 0.1–10 m M, but not at 100 m M (Fig. 2 *A*). Counts of viable cells for 4 d after removal of PHA showed a progressive increase in controls (without MTX), a borderline decrease in the presence of MTX at 0.01 and 100 m M, and a considerable loss of viable cells with MTX at either 1 or 10 m M (Fig. 2 *B*). When compared with measurements of apoptosis, these data indicate that MTX at low (0.01 m M) and high (100 m M) con-centrations inhibits cell proliferation without inducing apopto-sis, whereas at intermediate concentrations (from 0.1 to 10 m M), the decrease of viable cells is mostly accounted for by apoptosis. Addition of MTX at the onset of PHA activation resulted in a marked progressive decrease of viable cell counts in the presence of MTX (from 0.1 to 10 m M), whereas at low (0.01 m M) and m M) concentrations, MTX inhibited high (100

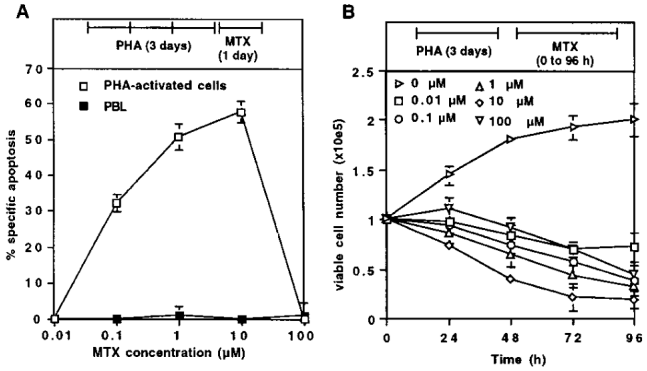

*Figure 2.*Characteristics of MTX-induced apoptosis. (*A*) Dose re-sponse: resting T cells (j) or cells activated by PHA (5 mg/ml) during 3 d (u) were treated with a dose range of MTX. Apoptotic cells were evaluated after 24 h. Values are the mean6SEM of three indepen-dent experiments. (*B*) Effect of MTX on viable cell recovery. PBL were stimulated with PHA (5 mg/ml) without or with MTX (10^{28} M to 10^{24} M) added at day 3. Viable cell number was determined by trypan blue exclusion at the indicated times. Values are the mean6SEM of three independent experiments.

and induced a transient decrease of viable cell numbers at in-termediate (0.1–10 m M) concentrations (data not shown). *MTX-induced apoptosis of activated T cells results from*

DHFR and TS inhibition. Knowing that MTX and its poly-glutamated derivatives can interfere with several folate-depen-dent enzymes, we determined whether blockade of the two major targets of MTX activity, DHFR and TS, could account for MTX-induced apoptosis of activated T cells. To this end, 3-d

PHA–activated cells were incubated with MTX (1 m M) to-gether with folic acid, folinic acid, or thymidine. Folinic acid showed dose-dependent inhibition of MTX activity, with z 80% decrease at m M (Fig. 4 *A*). Folic acid z 10–50 times less effective on a molar basis, and complete inhibition required 100 m M. Increasing the extracellular concentration of

*Figure 1.*MTX induces apoptosis of MTX). (*B* and *C*) **Morphology of BM quantum computers leverage furniture PBL after Hoescht 33342** staining and by electron microscopy, respectively (without MTX (*left*) or with MTX (*right*)). (*D*) Alteration of the mitochondrial transmem-brane potential (Dcm) measured by staining with DiOC6[3] and DNA strand breaks detection by the TUNEL assay as described in Meth-ods. Numbers in each histogram refer to the percentage of cells below (DiOC6[3]) or above (TUNEL) the threshold of normal controls.

cell proliferation but induced only a moderate and/or delayed loss of viable cells (data not shown).

PBL incubated with MTX undergo apoptosis upon subse-quent activation in drug-free medium. Preincubation of PBL with MTX (from 0.1 to 50 m M) resulted in apoptosis on subse-quent PHA activation for 24 h (Fig. 3 *A*), and maximal apopto-sis was attained after 8 h incubation with MTX (data not shown). To determine the half-life of MTX apoptosis–inducing

activity, PBL were incubated with MTX (1 m M) and, subse-quently, cultured in drug-free medium for various time inter-vals before activation by PHA. Results indicate a progressive linear decrease of the percentage of apoptotic cells with an av-

erage half-life of 3 d (Fig. 3 *B*). Counts of viable cells after PHA activation in drug-free medium demonstrated that MTX inhibits PHA-induced cell proliferation at 0.01 and 100 m M,

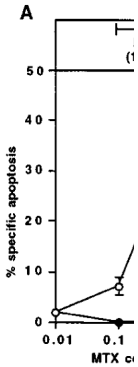

*Figure 3.*Apoptosis induced by MTX pretreatment: dose response (*A*) and half-life of intracellular MTX (*B*). PBL were

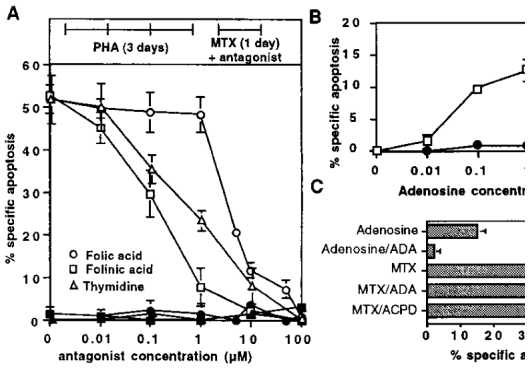

Figure 4. Effect of folic acid, folinic acid, or thymidine (*A*); adeno-sine deaminase and ACPD on apoptosis induced by adenosine or MTX (*C*). Resting T cells (*closed symbols*) or cells activated by PHA (5 mg/ml) in serum-free medium for 3 d (*open symbols*) were treated with MTX (10^{-6} M) plus a dose range of folic acid, folinic acid, or thy-midine (*A*) or with adenosine in serum-free medium. Identical values were observed in the presence of ACPD (10 mM) (*B*). 3-d PHA–acti-vated cells were incubated with ADA (2 U), or ACPD (10 mM), and treated 24 h with MTX (10^{-6} M) or adenosine (10^{-6} M) (*C*). Apop-totic cells were evaluated after 24 h, and results are expressed as per-cent-specific apoptosis as described in Methods. Values are the mean6SEM of three

cocultured with cells that express different MHC class II mole-

thymidine to 10 mM was sufficient to prevent 90% of MTX-induced apoptosis (Fig. 4 *A*).

Adenosine plays a limited role in MTX-induced apoptosis.

Knowing that adenosine represents the major mediator of the antiinflammatory activities of MTX (21), we investigated its contribution to MTX-induced apoptosis. Addition of adeno-sine (0.1–1 mM) in serum-free medium with or without ACPD resulted in apoptosis of z 15% of PHA-activated cells (Fig. 4 *B*). At higher concentrations (10–100 mM), adenosine in-duced some apoptosis of unactivated lymphocytes, but the per-centage of apoptotic PHA–activated cells did not increase. The apoptotic activity of adenosine (1 mM) was completely ab-rogated by adenosine deaminase, whereas that of MTX was only marginally decreased (Fig. 4 *C*). It was concluded that sensitivity to adenosine-mediated apoptosis was restricted to a subset of PHA-activated cells, accounting for z 15–20% of cells susceptible to MTX.

MTX-induced apoptosis requires progression to the S phase of the cell cycle. The apoptotic signal triggered by MTX is likely to be initiated at the time of DNA synthesis, during the S phase of the cell cycle. Supporting this hypothesis, the percent-age of blasts and the cell surface expression of CD69, CD25 (a

signaling receptor CD95 (Fas, APO-1), which mediates apop-tosis upon oligomerization. Because MTX was reported to stimulate CD95 ligand (CD95L) mRNA expression in human leukemia T cell lines (22), we studied the possible contribution of CD95L/CD95 interaction in the apoptotic effect of MTX. Expression of CD95L mRNA by 3-d PHA–activated cells could be readily induced by the association of a phorbol ester with a calcium ionophore, whereas MTX (10 and 200 mM) trig-gered only borderline CD95L gene transcription (Fig. 6 *A*). Apoptosis induced by MTX was not altered by addition of the CD95 antagonist antibody ZB4, which completely inhibits apoptosis induced by CD95L or by the CD95 agonist antibody CH11 (Fig. 6 *B*). PBL rapidly express CD95 upon activation, but their sensitivity to CD95-dependent apoptosis progres-sively develops from day 3 to day 6 of mitogenic activation, and requires IL-2 (23). After 24 h of mitogenic activation, PBL were fully susceptible to MTX-induced apoptosis (Figs. 3 *B* and 6 *C*), yet resistant to the CD95 agonist antibody CH11. Fi-nally, PBL from a patient with normal CD95 expression but defective CD95 signaling pathway, as shown by their resis-tance to CH11-induced

active T cells undergo activation and clonal expansion when

cules, a reaction termed MLR (mixed lymphocyte reaction) (24). Subsequent culture with the same stimulator cells results in accelerated proliferation, as shown by a peak of [^{3}H]TdR in-

corporation and CD25 expression at 2–3 d, whereas the re-

sponse to third party stimulator cells follows primary kinetics with maximal cell proliferation occurring after 5–6 d (13). Such differences indicate that the two types of stimulator cells are

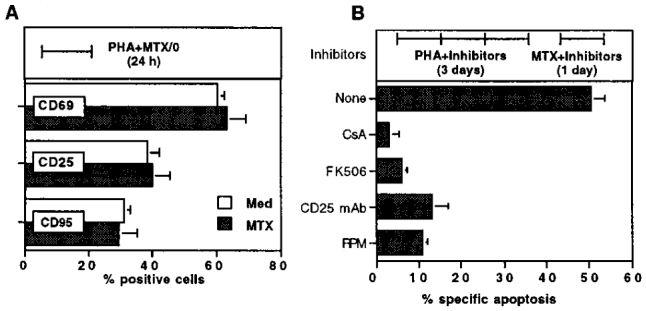

Figure 5. MTX-induced apoptosis and progression of activated PBL in the S phase of the cell cycle. PBL were activated by PHA (5 mg/ml) with or without MTX. (*A*) Percentage of cells expressing G1 phase markers (*CD69, CD25, CD95*) among PHA-activated PBL after 24 h in the presence (*gray bands*) or absence (*white bands*) of MTX (10^{-6} M). (*B*) Effect of inhibition of the G_1 to S phase transition on MTX-induced apoptosis. PBL were activated by PHA (5 mg/ml) with or without CsA (250 ng/ml), FK506 (10 nM), CD25 mAb (10 mg/ml), and RPM (60 nM). Results are expressed as the specific MTX apop-tosis, as described in Methods. Values are the mean6SEM of four in-dependent experiments. [^{3}H]TdR incorporation was determined dur-ing the last 8 h of culture with PHA: mean dpm (stimulation index in brackets). Nonactivated PBL: 300, PHA alone: 5,250 (17.5), PHA 1 CsA: 1,280 (4.2), PHA 1 FK506: 1,420 (4.7), PHA 1 CD25 mAb: 2,050 (6.8), PHA 1 RPM: 1,300 (4.3).

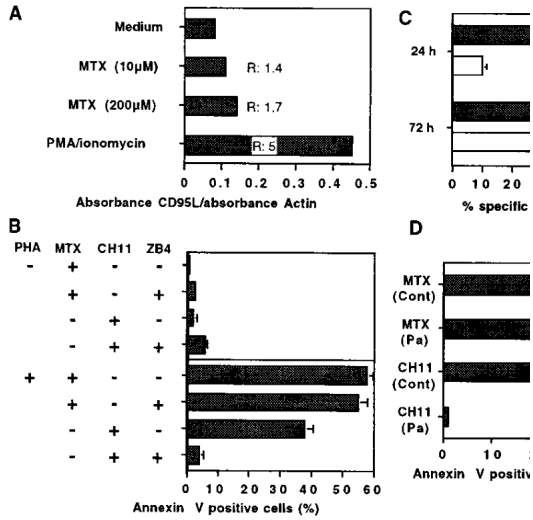

*Figure 6.*MTX-induced apoptosis does not involve CD95/CD95L in-teraction. PBL were cultivated for 3 d in the presence of PHA (5 mg/ml). Dead cells were eliminated by centrifugation on Ficoll, and viable cells were incubated with MTX (10 or 200 mM) or with PMA (10 ng/ml) plus ionomycin (0.5 mg/ml) for 12 h. mRNA of each sam-ple was amplified by reverse transcriptase–PCR as described in Methods, and separated on 2% agarose gel. Results are expressed as the ratio: absorbance of CD95L/absorbance of actin. *R* corresponds to the ratio: absorbance assay/absorbance of medium. (*B*) Effect of the antagonist Fas mAb ZB4 on MTX-induced apoptosis. 3-d PHA–activated cells were incubated 1 h with ZB4, and then treated with medium, MTX (10^{25} M), or the agonist CD95 mAb CH11 (1 mg/ml) for 24 h, and percentage of apoptotic cells was determined by annexin V binding. (*C*) Sensitivity to apoptosis induced by MTX and CD95 mAb CH11 according to culture duration. 1-d or 3-d PHA–activated cells were treated by MTX (10^{26} M) or CD95 mAb CH11 (1 mg/ml) for **apoptosis in lymphocytes from a patient BM quantum computers leverage furniture a CD95-signaling de-**fect. 3-d PHA–activated cells were treated with MTX (10^{26} M) or CH11 (1 mg/ml). After 12 h of incubation, the percentage of apoptotic cells was determined by flow cytometry after staining with annexin V.

upon subsequent mitogenic activation ex vivo (Fig. 8). In vitro apoptosis, determined after 24 h of activation by PHA, was completely inhibited in the presence of folinic acid (10 mM) and folic acid (100 mM) (data not shown), indicating that it could be attributed to MTX or its polyglutamated derivatives.

Discussion

MTX was shown to exert a wide range of antiinflammatory ac-tivities (21, 25–27) that are primarily mediated by the release of adenosine from different cell types that express ecto-59-nucleotidase (28). Such antiinflammatory properties do not ex-clude a genuine immunosuppressive activity, in as much as ADA deficiency is associated with a severe combined immu-nodeficiency disease (29). The present study demonstrates that MTX induces apoptosis of PBL activated by mitogens or su-perantigens, and may induce clonal deletion of alloreactive T cells at concentrations that are achieved during low-dose clini-cal treatments (30–32). Remarkably, nonactivated T cells were resistant to apoptosis at up to 1 mM MTX, demonstrating a se-lectivity toward activated PBL. Furthermore, resting PBL in-cubated with MTX, and subsequently activated in drug-free medium, underwent apoptosis, suggesting that unactivated T cells may convert MTX to polyglutamate forms that are re-tained intracellularly. The ability to undergo apoptosis may re-flect the capacity of PBL to convert MTX to MTX glu4 and MTX glu5, which were reported to be retained for up to 24 h in breast cancer cells (10, 11). In the present model, maximal apoptosis was achieved after 8 h of exposure to MTX and the half-life of the biological activity of polyglutamated MTX was around 3 d. Such information may be relevant for optimizing dosage and

recognized by distinct T cell clones, although cross-reactivity and/or HLA allele sharing between stimulator cells may often result in intermediate kinetics. To investigate whether a brief exposure to MTX could selectively delete activated T cells without impairing the functional capacities of other T cells in the same culture, we added MTX (1 mM) during the last 16 h of a primary MLR. Cells were then cocultured with either the same or a different allogeneic B cell line. In these conditions, the secondary response to iden-tical stimulator cells was nearly completely abrogated, whereas the response to a third party cell line was not affected (Fig. 7), indicating that the effect of MTX was clonally restricted to the T cells that were activated at the time of exposure to the drug.

*Lymphocytes from MTX-treated patients undergo apoptosis upon in vitro activation.*We verified in six rheumatoid arthri-tis patients, that a single injection of MTX during low-dose treatment was sufficient to prime lymphocytes to apoptosis

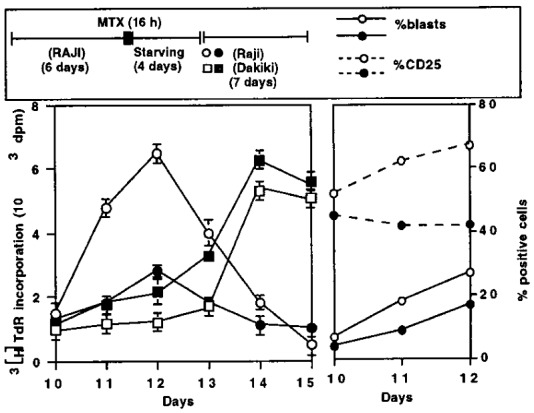

*Figure 7.*MTX-induced antigen–specific hyporesponsiveness in MLR. Lymphocytes from healthy adult donors were stimulated with RAJI cells with or without MTX (10^{26} M) added during 16 h at day 6 (*closed or open symbols,* respectively). Lymphocytes were washed and cultured in fresh medium from day 6 to 10, and then restimulated with RAJI or DAKIKI cells. [^{3}H]TdR incorporation in dpm (mean6 SD) and CD25 expression and blasts

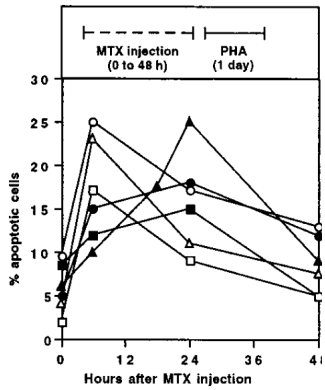

Figure 8. Ex vivo activation of PBL after MTX injection in-duces apoptosis. Six patients with rheumatoid arthritis un-der long-term treatment by MTX were studied. Blood samples were taken at indi-cated times after MTX (7.5–15 mg/kg) intramuscular injec-tion, and PBL were activated by PHA (5 mg/ml). Per-centages of apoptotic cells (Hoechst 33342 staining) were determined after 24 h PHA ac-tivation. Each symbol repre-sents one patient. Cultures in the presence of folic acid (100 mM) or folinic acid (10 mM) resulted in 1–5% specific apoptosis, as defined in Methods.

than that observed after in vitro incubation of normal PBL with MTX. Nevertheless, uptake and intracellular retention of MTX, along with the irreversibility of its effect on activated lymphocytes, provide a rationale for intermittent administra-tion in contrast with other antiinflammatory and immunosup-pressive agents that must be administered daily because of their short half-life and/or reversible activity.

Polyglutamated MTX not only blocks DHFR, the main tar-get enzyme of MTX, but also inhibits a number of folate-dependent enzymes, resulting in inhibition of purine and thymidylate synthesis (10, 11). Complete blockade of DHFR results in the depletion of intracellular reduced folates, but the apoptotic activity of MTX was only reversed at high molar ra-tios of folic acid to MTX, and much more efficiently by folinic acid, which bypasses DHFR. Complete inhibition by exoge-nous thymidine supports the prediction that this immunosup-pressive activity of MTX may be primarily controlled by intra-cellular levels of thymidine.

Adenosine accounted for only a minor part of MTX-induced apoptosis because adenosine-triggered apoptosis was re-stricted to a small subset (z 15%) of activated PBL. As dis-cussed below, programmed cell death of activated PBL after MTX treatment was initiated at

is of major relevance for therapeutic applications because CD95-dependent apoptosis is restricted to a subset of preacti-vated "memory" T cells that express the CD45 RO short iso-form (37–39), and that have developed sensitivity to the CD95-mediated apoptosis pathway through an efficient IL-2 signal (23). Conversely, MTX induces apoptosis of naive as well as memory T cells, and may induce deletion of naive activated T cells, as shown by MLR experiments.

MTX is usually used as synergy between MTX and drugs that interfere with the IL-2 pathway, unless MTX deletes T cells that have escaped activa-tion blockade by the associated drugs. This could be the case in autoimmune disorders where CsA is administrated at much lower doses than in transplantation.

In conclusion, the present study demonstrates that low-dose MTX treatment can induce apoptosis and clonal deletion of activated T cells, and, thus, establishes the potent immuno-suppressive properties of this drug. The model provides a ra-tionale for intermittent administration and it could be applied to optimizing the pharmacokinetics of MTX treatment. How-ever, the relative irreversibility of this effect may represent a risk of overimmunosuppression in long-term therapy.

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